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Time-lapse Monitoring of Pressure Depletion for High-Temperature Gas-Condensate Reservoirs to Optimize In-Fill Perforation Using Pulsed-Neutron Well Logging

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Abstract

When infill perforation intervention opportunities are evaluated, it is essential to confirm reservoir pressure. Without appraisal of reservoir pressure, perforation of depleted reservoirs can negatively impact the overall well performance and even introduce early water breakthrough. We successfully evaluated the reservoir pressures of unperforated high-pressure, high-temperature (HPHT) gas-condensate-bearing reservoir sands using pulsed-neutron well logging technology. This paper discusses multiyear time-lapse pulsed-neutron well logging in an HPHT gas-filled borehole using previous- and next-generation sleeved-multidetector instruments for through-casing reservoir pressure depletion monitoring.

We employed a previous-generation sleeved-pulsed-neutron tool to acquire three-detector-based pulsed-neutron capture (PNC) data through a gas-filled wellbore to analyze reservoir pressure depletion in 2022. The challenge in the first logging run was a high-temperature environment close to the tool operating limit. Surveillance revealed that the upper target sand was not depleted, whereas the lower sands were depleted. The operator planned to perform time-lapse monitoring of the target sands in 2024 before the final perforation decision. The next-generation sleeved-pulsed-neutron tool was deployed in the 2024 run. PNC data were recorded twice as fast as the first run using the legacy tool.

Crucial technological advancements in the next-generation pulsed-neutron tool enable time-efficient PNC data acquisition. First, the pulsed-neutron generator emits twice as many high-energy neutrons as the legacy tool. Second, the adoption of denser scintillation material for gamma-ray detectors allows for recording higher gamma-ray count rates with higher resolution. Finally, a high-speed digital electronics system processes gamma rays faster. The logging speed of the multidetector PNC mode with the new tool was twice that of the legacy tool at 20 ft/min. Benefiting from a faster logging speed, thus reducing the time the tool is in the hole, the heat in the borehole, approximately 350 degrees Fahrenheit, was not a significant concern. The data obtained using the next-generation pulsed-neutron tool demonstrated reliable coherence with the previous data acquired via the legacy tool, confirming a smooth transition from the old tool to the new tool and ensuring seamless time-lapse data analysis. Additionally, extensive Monte Carlo N-Particle (MCNP) models under various reservoir pressure conditions for legacy and new tools were integrated into

the reservoir depletion analysis. Time-lapse analysis revealed that the upper sand remained under the virgin reservoir condition but that the lower sands unconnected to the upper sand underwent reservoir depletion.

The target reservoir was subsequently perforated, confirming the virgin pressure status as predicted by the pulsed-neutron logging appraisal. The single-well opportunity increased productivity by approximately 15,000 barrels of oil equivalent per day (BOE/D). Technological advancements, resulting in increased logging speeds, extended the operator's operating window at high temperatures, allowing for a greater appraisal scope. Cased-hole depletion analysis presents a low-risk, cost-effective screening tool for in-fill production opportunities in late-life field production.

Introduction

Openhole well logging and analysis of acquired datasets, including natural radioactive gamma-ray, neutron porosity, bulk density, and electrical resistivity logs, are commonly used to define gas-bearing reservoirs and assess their gas saturation. Once factors such as the geological connectivity of reservoirs to offset wells, potential recoverable gas volumes, and the management of unwanted fluid production are reviewed, perforation of hydrocarbon-bearing zones typically proceeds. A crucial condition to confirm before perforating the zones in a gas-bearing reservoir is whether the current reservoir pressure is similar to or close to the original reservoir pressure. The reservoir pressure of target gas-bearing zones must be assessed. Therefore, zones exhibiting reservoir depletion are not perforated because depleted reservoirs may not yield gas production as anticipated, as initially indicated by openhole well logs, and may significantly increase water cut adversely. Furthermore, perforating depleted gas reservoirs, especially in offshore fields, leads to unfavorable operational and economic conditions.

A non-destructive method to evaluate reservoir pressure depletion through cased wellbores is enabled using pulsed-neutron well logging technology. We adopted multidetector, multi-functional pulsed-neutron well logging instruments to record pulsed-neutron capture (PNC) data. We then analyzed the acquired data by integrating forward models of pulsed-neutron measurements with prior knowledge of the reservoir, including fluid volumes under virgin pressure conditions, as well as the volumes of sand and shale.

We implemented this reservoir pressure surveillance technology for offshore high-pressure, high-temperature (HPHT) gas-condensate reservoirs located in the North Sea. Pulsed-neutron well logging was conducted in 2022 and 2024. We used first-generation and next-generation multidetector slim-hole pulsed-neutron well logging tools for the two respective monitoring logging surveys. The time-lapse monitoring of the targeted gas-condensate reservoirs significantly enhanced the operator's understanding of these reservoirs, supporting informed decision-making regarding the infill perforation strategy for enhancing gas production.

Pulsed-Neutron Well Logging Technology

First-Generation Multidetector Pulsed-Neutron Logging Tool

The advancement of technology in pulsed-neutron well logging tools has led to the introduction of the industry's first slim-hole, 1.69-inch outer-diameter, three-scintillation-detector pulsed-neutron well logging tool in 1999 (Gilchrist et al., 1999). The three gamma-ray detectors are referred to as short-, long-, and extra-long-spaced detectors, respectively, based on their proximity to the pulsed-neutron generator. This multidetector pulsed-neutron well logging tool has multifunctional characteristics, enabling the acquisition of diverse pulsed-neutron datasets, including PNC, carbon/oxygen (C/O) ratios (Kim et al., 2023c), and oxygen activation gamma-ray count rates for assessing water movement (Chace et al., 1994; Kim et al., 2022), as well as pulsed-neutron holdup measurements (Trcka et al., 1996; Kim et al., 2022). Moreover, the launch of this pulsed-neutron well logging tool has accelerated the development of answer products using the acquired data, including two-phase and three-phase saturation analyses (Cedillo et al., 2015;

Itter et al., 2015; Conner et al., 2016; Kim et al., 2013; 2016; 2017; 2021; 2022; 2023a; 2024a; 2024b; 2025). Among the various answer products, the quantification of gas saturation through gamma-ray ratio measurements emerged, and the primary principle for analyzing reservoir pressure depletion is based on this gas quantification method (Treka et al., 2006; Kim et al., 2023d).

Next-Generation Multidetector Pulsed-Neutron Logging Tool

The most recent enhancements to pulsed-neutron well logging tools include (1) an increased capacity for the pulsed-neutron generator, (2) a high-resolution, high-density scintillation gamma-ray detector material, and (3) upgraded gamma-ray processing electronics (Nardiello et al., 2022; Kim et al., 2023a; Akagbosu et al., 2023; and McGlynn et al., 2023). The latest pulsed-neutron tool can generate twice as many fast neutrons as its predecessor. The addition of a lanthanum bromide (LaBr_3) crystal in the scintillation detector allows for higher gamma-ray count rates and improved precision in energy spectral data acquisition. Furthermore, the use of digital electronics reduces dead time, enabling more efficient gamma-ray processing.

The advancement of these three main components allows for the acquisition of multidetector pulsed-neutron capture data (which are used for reservoir depletion surveillance analysis) with the next-generation pulsed-neutron well logging tool at twice the rate of the legacy tool. The multidetector PNC data can be logged at a speed of 20 ft/min with the next-generation pulsed-neutron tool, compared to 10 ft/min with the previous-generation tool.

The acquisition speed is a critical factor in logging the HPHT reservoirs. The operating temperature limit of pulsed-neutron well logging instruments is 175 degrees Celsius. The borehole temperature for the subject well is close to the pulsed-neutron's operating temperature limit. Therefore, the use of the next-generation pulsed-neutron well logging tool is beneficial to minimize the duration of the pulsed-neutron well logging tool in the borehole, ensuring that high temperatures do not compromise the recorded data.

Sleeved-Pulsed-Neutron Well Logging Technology

The ideal borehole condition for pulsed-neutron well logging occurs when it is filled with water. Such a water-filled borehole, characterized as a hydrogen-sufficient environment, promotes the transverse propagation of bombarded neutrons toward the formation, thereby inducing characteristic gamma rays by elements in the formation. Conversely, a gas-filled borehole, characterized by a hydrogen-deficient condition surrounding the pulsed-neutron tool, leads to a detrimental suppression of formation sensitivity in pulsed-neutron measurements (Zhang et al., 2017; Raesi et al., 2017; and Kim et al., 2023b).

The solutions to this unfavorable pulsed-neutron well logging condition have been identified as establishing hydrogen-abundant borehole conditions through the injection of water prior to and during the pulsed-neutron well logging operation or by encasing the pulsed-neutron tool with a hydrogen-rich material. Gas field operators have preferred the latter solution, as sleeved-pulsed-neutron logging mitigates potential health, safety, and environmental (H&SE) issues associated with water injection, as well as damage to adjacent perforation zones and the formation. As a result, we used 3.5-inch outer diameter sleeved-pulsed-neutron well logging tools for both surveillance operations.

Time-Lapse Monitoring Using Two Distinct Pulsed-Neutron Well Logging Tools

Background

The subject well intersects two primary gas-condensate-bearing sand formations, and the maximum well deviation is approximately 44 degrees. Each formation comprises several pay sand bodies, with average shale volumes ranging from approximately 10 to 30 percent. The deeper sand formation has been in production from the subject well, while the upper sand formation was undergoing evaluation of depletion prior to the perforation decision.

Several wells access the upper targeted gas-condensate-bearing sandstone formation. Formation pressure testing, conducted in 2020 under openhole conditions from the adjacent well for the target formation, showed a decrease in reservoir pressure, indicative of potential reservoir depletion. The other offset well, which has perforations across the target formation, primarily contributed water. Monitoring the target reservoir pressure from the subject well was crucial prior to creating perforations in this wellbore.

The initial monitoring of the offshore HPHT well was conducted in 2022 using a previous-generation pulsed-neutron well logging tool. During the first logging survey, challenges were encountered pertaining to the high-temperature environment. Consequently, a subsequent survey was conducted in 2024 using the next-generation pulsed-neutron tool, which offered a faster logging speed than the legacy tool, thereby mitigating the unfavorable temperature conditions.

A significant consideration in employing two distinct logging instruments for time-lapse reservoir monitoring, utilizing identical log measurements, pertains to the compatibility of data acquired from both previous- and next-generation tools. Kim et al. have documented the compatibility of data obtained through these two instruments, elaborating on the criteria for data acquisition, the precision of recorded data, and the outcomes of post-processing data analysis (Kim et al., 2023a).

Stochastic Nuclear Modeling for Previous- and Next-Generation Tool Responses

A comprehensive Monte Carlo N-Particle (MCNP) modeling has been conducted to predict two gamma-ray ratio measurements derived from previous- and next-generation sleeved-pulsed-neutron tools. The ratio of gamma-ray count rates from the short- and extra-long-spaced detectors, which are extracted from the inelastic time domain, is defined as RIN13. Conversely, the same ratio measurement from the capture time domain in the time spectra is denoted as RATO13. Inanc et al. and Kim et al. have thoroughly discussed the characteristics of these two ratio measurements and compared their strengths and limitations (Inanc et al., 2009; Kim et al., 2022, 2023d)

Fiberglass sleeve-equipped legacy and next-generation pulsed-neutron tools were used, enhancing the formation sensitivity of the measurements while logging in a gas-filled borehole, thereby minimizing uncertainty in depletion analysis. Consequently, RIN13 and RATO13 responses from two sleeved-pulsed-neutron tools were modeled. Table 1 shows the MCNP modeling parameters for the upper sandstone formation.

Table 1—MCNP modeling parameters for time-lapse reservoir depletion analyses for the upper sandstone formation

Surveillance year	2022	2024
Pulsed-neutron tool type	Sleeved-previous-generation	Sleeved-next-generation
Hole size	8.5 inches	
Casing size	5 inches (outer diameter), 4.044 inches (inner diameter)	
Lithology	Sandstone	
Annular cement condition	Well-cemented condition	
Borehole fluid type and density	Gas and 0.23 g/cc	Gas and 0.17 g/cc
Formation water salinity and density	240,000 ppm and 1.1 g/cc	
Formation condensate density	0.55 g/cc	
Formation gas density	0.15, 0.2, 0.25, and 0.332 g/cc	0.15, 0.2, 0.25, and 0.332 g/cc
Reservoir pressure corresponding to the estimated gas density	3,700, 5,350, 7,400, and 13,450 psi	3,700, 5,350, 7,400, and 13,450 psi

Fig. 1 illustrates the estimation of in situ formation gas density as a function of reservoir pressure for the upper sandstone formation, represented by circles, at a reservoir temperature of 170 degrees Fahrenheit (Fig. 1(a)), as well as for the deeper sandstone formation, indicated by rectangles, at a reservoir temperature

of 178 degrees Fahrenheit (Fig. 1(b)). As the reservoir pressure escalates, the density of the gas swiftly increases to 0.25 g/cc. Subsequently, the rate of increase in gas density diminishes as the reservoir pressure approaches the virgin pressure. Moreover, the gas density profiles observed in the upper and lower sand formations show notable similarities.

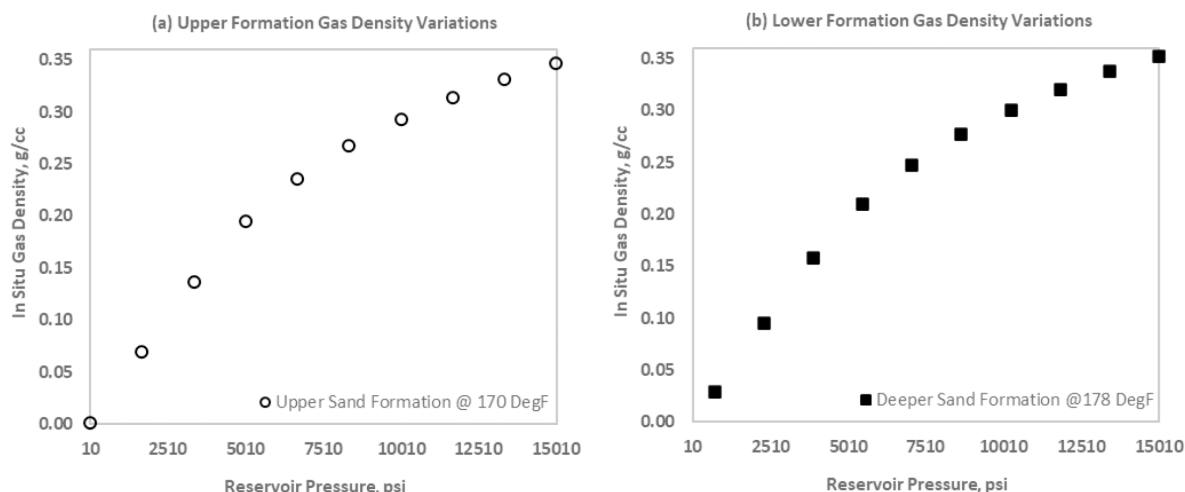


Figure 1—Estimated in situ formation gas density with respect to reservoir pressure and temperature for two sandstone formations: (a) upper sand formation case and (b) lower sand formation case

The MCNP models for RIN13, produced by both previous- and next-generation sleeved-pulsed-neutron well logging tools, are shown in Fig. 2(a) and Fig. 2(b), respectively. The horizontal axis indicates porosity, expressed in fractions, while the vertical axis represents RIN13. Each line represents formations filled with a specific formation fluid in a defined logging environment (Refer to Table 1). For instance, the blue line was created under the condition where the formation is fully saturated with 1.1 g/cc water. Conversely, the red line represents the formation fully saturated with 0.332 g/cc gas (i.e., the initial gas density estimated with the virgin reservoir pressure). It is noteworthy that the RIN13 envelopes presented in Fig. 5 are based on these MCNP models, incorporating formation volumetrics.

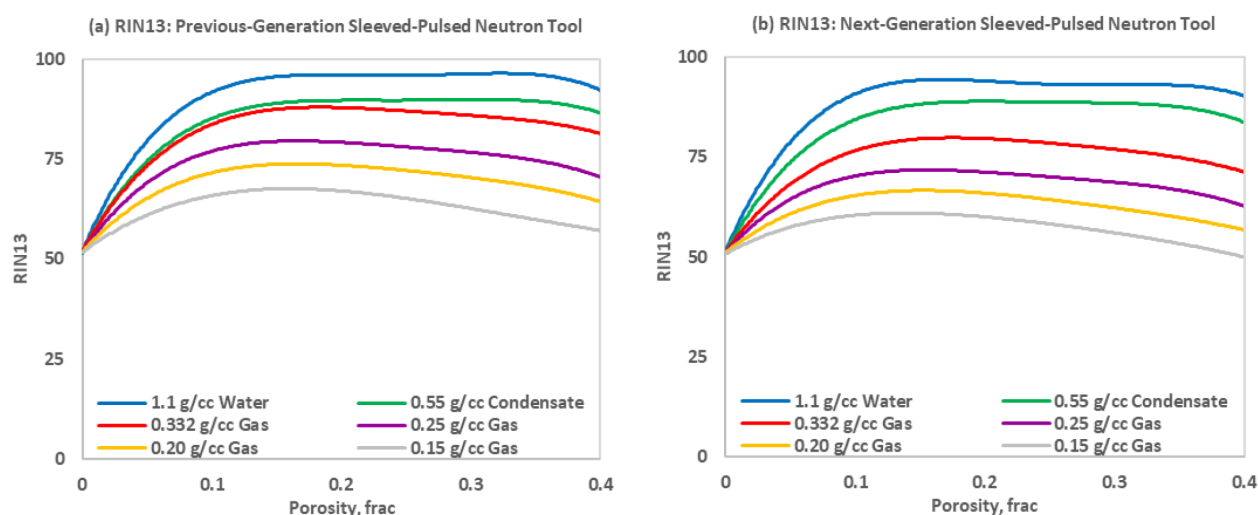


Figure 2—MCNP modeling results for RIN13 of (a) previous- and (b) next-generation sleeved-pulsed-neutron well logging tools under various formation fluids, including variations in gas density. RIN13 is sensitive to variations in reservoir pressure and is used to analyze reservoir pressure depletion.

The RIN13 responses for all fluids exhibit similar profiles, characterized by a rapid increase in lower porosity and either a stable or gradual decrease in higher porosity. RIN13 for the next-generation tool is lower than that for the previous-generation tool. Comparisons between the responses of the previous- and next-generation pulsed-neutron tools in a field trial project for the next-generation tool revealed that the RIN13 difference between the two tools under the same logging conditions was minimal (Kim et al., 2023a). Therefore, the difference shown in Fig. 2 is attributed to the lower gas density during the second surveillance run compared to the first run.

The RATO13 MCNP models are presented in Fig. 3. The RATO13 model for 1.1 g/cc water-filled formation is clearly distinguished from the other RATO13 models for hydrocarbon-filled formations across all levels of porosity. This distinction exists because RATO13 is influenced by water salinity, demonstrating a positive correlation with it. The wider dynamic range between water- and gas-filled formations, primarily shaped by the saline water environment, provides better resolution compared to that of the RIN13. However, RIN13 demonstrates greater sensitivity to variations in gas density, which is a fundamentally essential characteristic for depletion analysis (Nardiello et al., 2017; Kim et al., 2023d). In comparison to formation sigma, RATO13 exhibits increased sensitivity to gas density variation.

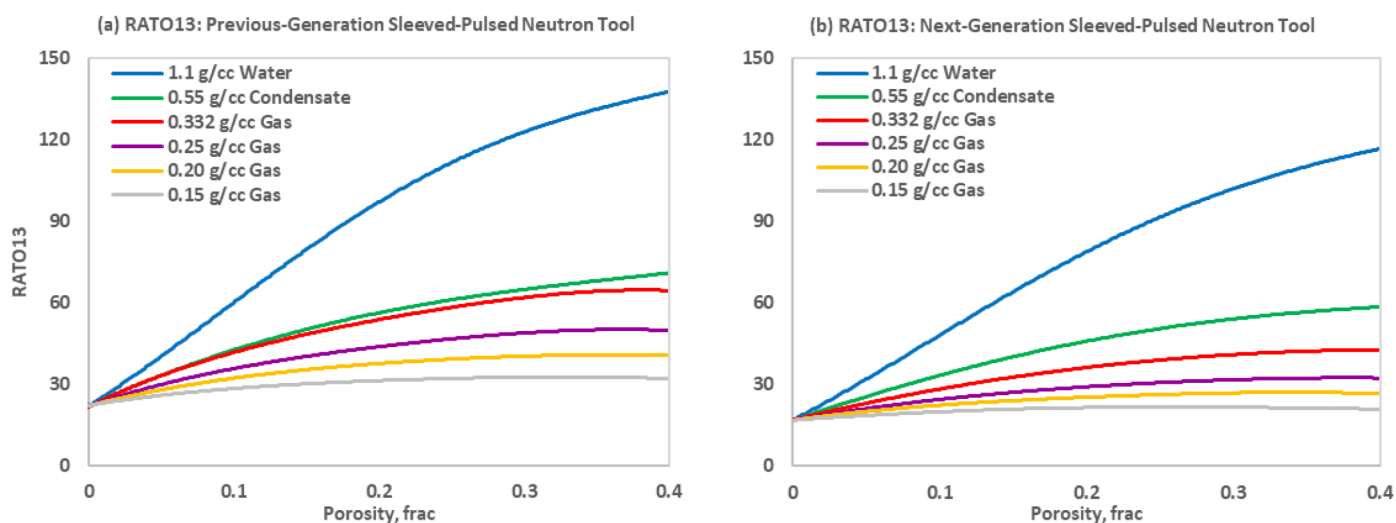


Figure 3—MCNP modeling results for RATO13 of previous- and next-generation sleeved-pulsed-neutron well logging tools under various formation fluids, including variations in gas density. RATO13 is less sensitive to variations in reservoir pressure than RIN13 but has a greater dynamic range between water- and gas-filled formations.

As porosity increases, a divergent trend is observed between RIN13 and RATO13. Specifically, RATO13 continues to show an upward trend with the increase in porosity. Conversely, as the density of the formation gas decreases, the RATO13 responses do not show the same increase as observed in RATO13 responses for formations with higher fluid density.

Time Comparative Reservoir Pressure Depletion Analyses

The Depletion Analysis Algorithm

The depletion analysis applies to the non-perforated interval, where the original formation fluids remain unaltered. The reverse engineering of the forward model-based gas saturation analysis constitutes the fundamental principle (Trcka et al., 2006; Ansari et al., 2009; Nardiello et al., 2017; Gray et al., 2019; and Kim et al., 2023d). An iterative comparison of the forward-modeled responses of RIN13 and RATO13, along with their respective measured data, is necessary. If the measured data are lower than the predicted forward-modeled responses based on the virgin reservoir pressure, then the depletion of reservoir pressure is confirmed.

Given the nature of the depletion analysis algorithm, it is essential to generate precise MCNP models to develop theoretical tool responses for RIN13 and RATO13. Moreover, understanding the relationship between pulsed-neutron measurements used in the analysis and various near-wellbore logging environments, such as differences in wellbore fluid density, the quality of the annular cement sheath, and heterogeneity in formation lithology and porosity, is crucial (Kim et al., 2023b).

Specifications of Logging Environments

The specifications of the two pulsed-neutron well logging surveys are detailed in Table 2.

Table 2—A comparison table presenting specifications of the two pulsed-neutron logging operations

Surveillance year	2022	2024
Well condition	Shut-in	
Pulsed-neutron tool type	Previous-generation sleeved	Next-generation sleeved
Logging speed	10 ft/min	20 ft/min
Borehole fluid density (derived using pressure measurement)	Approximately up to 0.23 g/cc	Approximately up to 0.17 g/cc
Borehole temperature	315–318 degrees Fahrenheit (157–159 degrees Celsius)	327–342 degrees Fahrenheit (164–172 degrees Celsius)
Pulsed-neutron tool temperature	350–360 degrees Fahrenheit (177–182 degrees Celsius)	345–357 degrees Fahrenheit (174–181 degrees Celsius)
Borehole pressure	5,360–5,425 psi (370–374 bars)	3,654–3,872 psi (252–267 bars)

In both instances, the well was maintained in a shut-in condition. All other parameters exhibited minor variations between the two logging runs. The increased logging speed observed in the second run was achieved through the deployment of advanced, next-generation tools. A comparison of the standard deviation values of RIN13, RATO13, and formation sigma measurements between the two runs indicated that the data quality obtained during the second run was uncompromised. Borehole temperature sensors, deployed in conjunction with the pulsed-neutron tool, showed that the ranges of borehole temperatures exhibited less than a 10% variance. The recorded temperature of the tool remained close to the operational temperature specifications. The pressure sensor, which measured borehole pressure, revealed a reduction in pressure of approximately 30% in the second run. The fluid density, estimated using the measured pressure values, indicated a gas density disparity of approximately 0.05 g/cc (i.e., 0.23 g/cc in 2022 versus 0.17 g/cc in 2024). This decrease in borehole pressure may be attributed to a reduction in the reservoir pressure of the perforated deeper sand.

Time-Lapse Logging Analysis and Interpretation

Raw Data Comparison. Fig. 4 presents crossplots of RIN13, RATO13, formation sigma, pulsed-neutron porosity derived from time-lapse PNC data, borehole pressure, and borehole temperature.

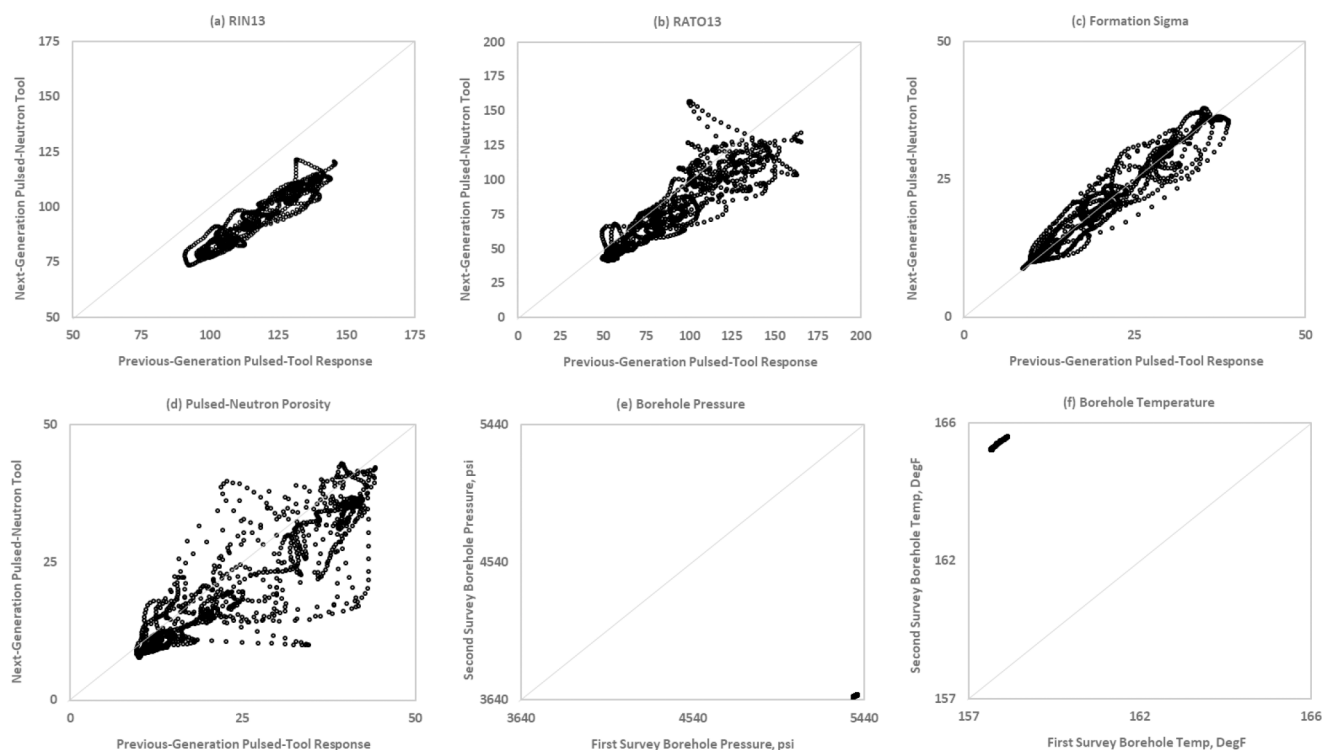


Figure 4—Crossplots of four key measurements obtained using both previous- and next-generation pulsed-neutron tools for the upper sandstone formation, encompassing upper and lower sand bodies: (a) RIN13, (b) RATO13, (c) formation sigma, (d) pulsed-neutron porosity, (e) borehole pressure, and (f) borehole temperature.

The RIN13 data exhibit a scatter below the unity slope, consistent with the MCNP modeling results presented in Fig. 2. Notably, RIN13 values obtained from the previous-generation pulsed-neutron tool surpass those from the next-generation tool. Additionally, the RATO13 crossplot reveals that values recorded with the previous-generation tool exceed those obtained with the next-generation tool. It is important to note that the observed decrease in values is primarily due to the reduced gas density in the borehole between the first and second surveys. Specifically, the gas density in the borehole decreased from 0.23 to 0.17 g/cc, resulting in a corresponding reduction in hydrogen density, which ultimately led to lower values in RIN13 and RATO13, respectively. The time-lapse formation sigma data align along the unity slope, indicating no significant change in formation water volume. The pulsed-neutron porosity data also exhibit some scatter in relation to the unity slope; however, certain widely scattered data points are affected by the reduction of wellbore fluid density. Time-lapse borehole pressure and temperature were crossplotted and showed that borehole temperature was higher, while borehole pressure was lower in the second surveillance run.

A Composite Log Plot. Fig. 5 presents a composite log of time-lapse pulsed-neutron logs and auxiliary logs. Each horizontal grid line represents 10 meters.

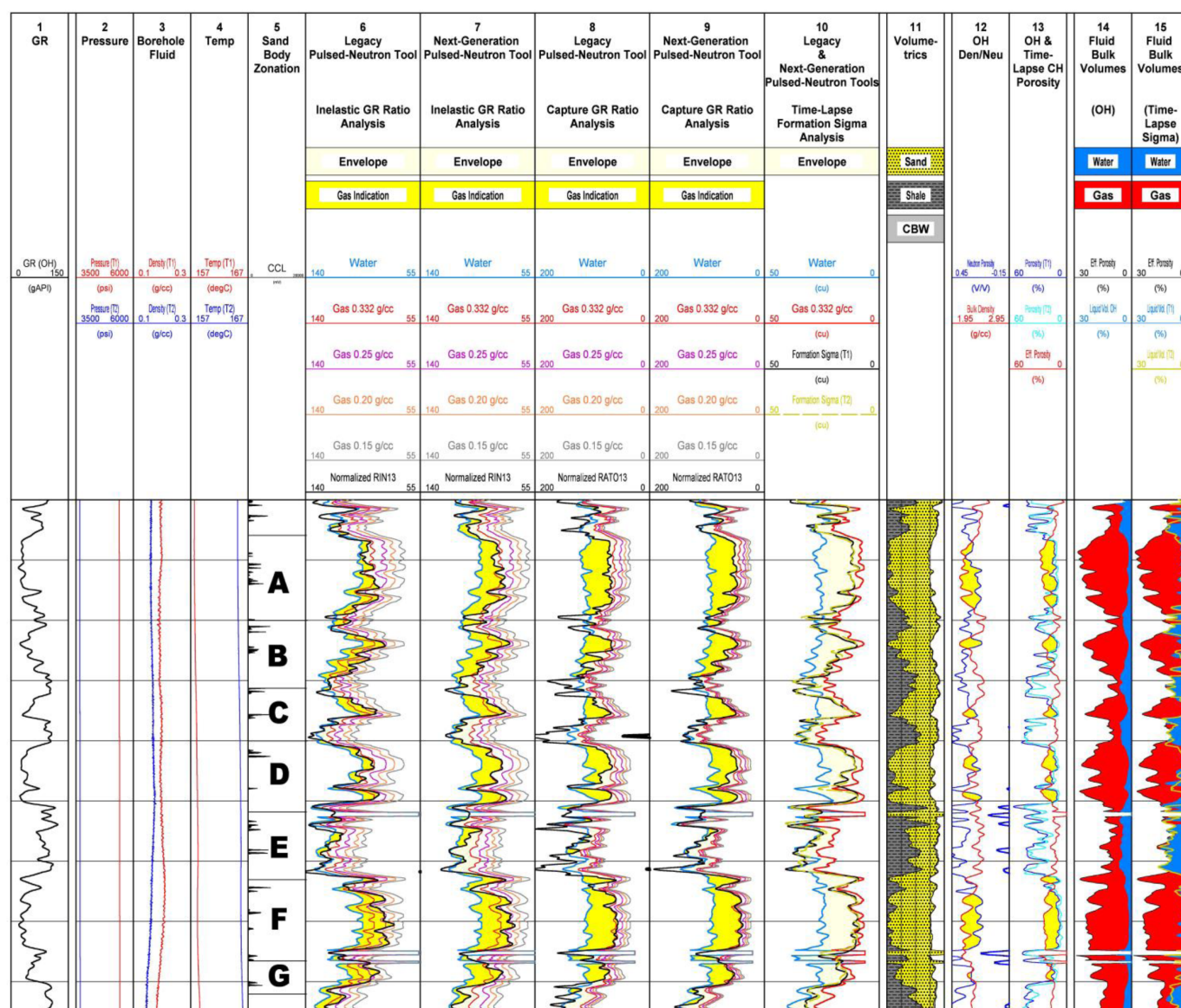


Figure 5—A composite time-lapse pulsed-neutron well logging data analysis plot for assessing reservoir depletion and saturation.

Track 1 includes a gamma-ray log obtained under openhole conditions. Track 2 displays borehole pressure measurements: the first run is highlighted in red, while the second run is shown in blue. Borehole pressure decreased by approximately 1,700 psi in the second run. Fluid density, estimated based on the measured pressure data, included a deviation correction. The corresponding pseudo-density logs are presented in Track 3. Track 4 features time-lapse borehole temperature logs, indicating that the second run recorded higher borehole temperatures. In summary, the borehole conditions from the second survey exhibited lower pressure but higher temperature and lower-density gas compared to the borehole conditions of the first survey.

Track 5 presents casing collar locator (CCL) log and sand bodies from 'A' to 'G' in the upper sandstone formation.

Tracks 6 and 7 illustrate theoretical RIN13 responses based on the volumetrics shown in Track 11 and the measured RIN13 from the two pulsed-neutron tools. Similarly, Tracks 8 and 9 present RATO13 envelopes along with measured data for both tools. Each track's MCNP model-based theoretical tool response considers water-filled formations and various gases representing conditions from virgin reservoir

pressure to depleted formations. When the formation is saturated with water under specified volumetric conditions, the RIN13 or RATO13 response, depicted in blue, reflects the water side of the envelope. In contrast, when the formation is saturated with gases of varying densities, the RIN13 or RATO13 responses (shown in red, purple, orange, and gray) illustrate the gas side of the envelope. The measured RIN13 or RATO13 is displayed in black. The borehole fluid difference between the two surveys was eliminated, as the MCNP modeling already accounts for this discrepancy. The relative position of RIN13 or RATO13 within their respective envelopes signifies the degree of reservoir depletion.

As mentioned earlier in this paper, the RIN13 measurement is more sensitive to variations in gas density than the RATO13 measurement and is therefore generally preferred for depletion analysis. However, in the case of logging across the gas-filled wellbore, RIN13 exhibited a relatively smaller envelope size, primarily attributable to the presence of gas within the wellbore, although the sleeved-pulsed-neutron tool occupied some additional space inside the gas-filled casing. The envelope sizes shown in RIN13 analysis tracks are narrower than those in RATO13 analysis tracks.

For formation sigma envelopes, the use of MCNP models is not required. Instead, a material balance equation is employed, which incorporates the volumes of sand and shale, as well as the corresponding Sigma water and gas values across various gas densities. Formation sigma-based envelope and time-lapse data are presented in Track 10. Multiple gas lines were excluded from the sigma analysis due to the inadequacy of sensitivity variations in gas density for conducting depletion analysis. The formation sigma log, depicted in black, was recorded using the previous-generation tool in 2022, whereas the log represented in yellow was obtained with the next-generation tool in 2024. The formation fluid volumes calculated from the formation sigma logs are illustrated in Track 15. The shaded water and gas indications are derived from the initial survey, while the yellow log reflects the water-filled porosity from the second survey analysis. The water saturation profiles have exhibited consistency over the years. In conjunction with the initial reservoir conditions, the hydrocarbon saturation profile presented in Track 14 demonstrates that the water saturation has remained unchanged since the acquisition and evaluation of the openhole logs.

Finally, time-lapse pulsed-neutron porosity logs are compared in Track 13 and overlaid with effective porosity. The separation between the effective porosity and the pulsed-neutron porosity from the first run was shaded in yellow, indicating the presence of formation gas. The qualitative indication of gas presence showed good agreement with the one observed from openhole gas indication in Track 12.

Log Interpretation.

Upper Sand Layers from 'A' to 'E'

The analysis of the RIN13 envelope and the measured data for the upper sands (from 'A' to 'E') indicates that both surveillance analyses corroborate the absence of any evident indications of reservoir pressure depletion. In both instances, the measured RIN13 is generally situated within the respective gas side of the envelope, which is illustrated in red. This gas line has been established based on a virgin gas pressure of 13,450 psi, corresponding to a density of 0.335 g/cc. Regarding the same sand bodies, the RATO13 measurements obtained from the two surveys fall within the RATO13 envelope constructed using 1.1 g/cc water and 0.335 g/cc gas lines.

The water saturation analysis, employing formation sigma, demonstrates strong concordance with the openhole resistivity-based saturation profile. In addition to calculating water saturation based on formation sigma, the overlay of effective porosity and pulsed-neutron porosity demonstrated that the hydrocarbon (or water) saturation profiles from the 2022 and 2024 surveillance remained consistent. The observed variance in scales between the two cased-hole porosity logs was attributed to differences in borehole fluid properties. Furthermore, a general concordance of hydrocarbon indication was noted via the overlay of openhole bulk density and neutron porosity measurements.

In summary, the log interpretation of eight log measurements (two RIN13, two RATO13, two formation sigma, and two pulsed-neutron porosity logs) concludes that the upper sands remain under original reservoir pressure conditions and contain hydrocarbons similar to the original condition.

Lower Sand Layers from 'F' to 'G'

Responses of RIN13 and RATO13 for the lower sand bodies (Sands 'F' and 'G'), as compared to the upper sand bodies, indicated that the two lower sands were depleted in 2022 and remain in a similar depleted state in 2024. The RIN13 logs measured in 2022 and 2024 were lower than the original gas line (i.e., the red line generated with a gas density of 0.332 g/cc) and were close to the gas line constructed with a gas density of 0.25 g/cc. The corresponding reservoir pressure was approximately 7,400 psi, suggesting a 45% reduction from the initial reservoir condition. The RATO13 logs for both runs were positioned on or slightly below the virgin gas density-based theoretical gas line (i.e., the red RATO13 line). Compared to the reservoir depletion indication using RIN13, the indication from RATO13 was less pronounced due to its lower sensitivity to variations in gas density, which represents reservoir pressure. Nonetheless, RATO13 responses, in relation to theoretical (or forward-modeled) responses for the lower two sands, were relatively lower than those for the upper sands. Although the sands were depleted, the quantified gas in the formation, computed using formation sigma, was similar to the original condition.

Reservoir Depletion Analysis Summary

The first pulsed-neutron well logging analysis indicated the reservoir depletion for the lower sand bodies, whereas the upper sands remained in a virgin pressure condition. The 2024 data analysis across the unperforated target formation indicated that no significant changes in reservoir pressure were observed between the two surveys.

The two surveillance analyses that evaluate reservoir depletion provided the operator with reliable guidance for optimizing in-fill perforation opportunities, consequently maximizing hydrocarbon recovery. As a result, the operator has perforated non-depleted sand bodies, resulting in an increase in production of up to 15,000 barrels of oil equivalent per day (BOE/D) from the subject well.

Extending Surveillance to the Deeper Formation

The deeper sandstone formation was perforated and produced hydrocarbons. The evaluation of the remaining hydrocarbon volume in the producing reservoirs is of interest to the operator. In the pre-job planning stage for the first pulsed-neutron well logging in 2022, the logging program was structured to incorporate data acquisition for the deeper sandstone formation, contingent upon the tool's resilience to the high temperatures encountered while logging the upper sandstone formation. During the first survey with the previous-generation pulsed-neutron tool, PNC data recording was confined to the upper sandstone formation as the tool was operating near its maximum capacity. Conversely, the second survey, employing the next-generation pulsed-neutron tool, achieved data acquisition at twice the speed of the first survey, enabling the logging of the deeper sand formation, despite the same temperature limitation as the legacy tool. The improvement and advancement of pulsed-neutron logging technology allowed the reservoir surveillance envelope to extend further.

Pulsed-neutron log analysis was performed to delineate the saturation profile of the producing formation, as shown in [Fig. 6](#). One horizontal grid line indicates 20 meters.

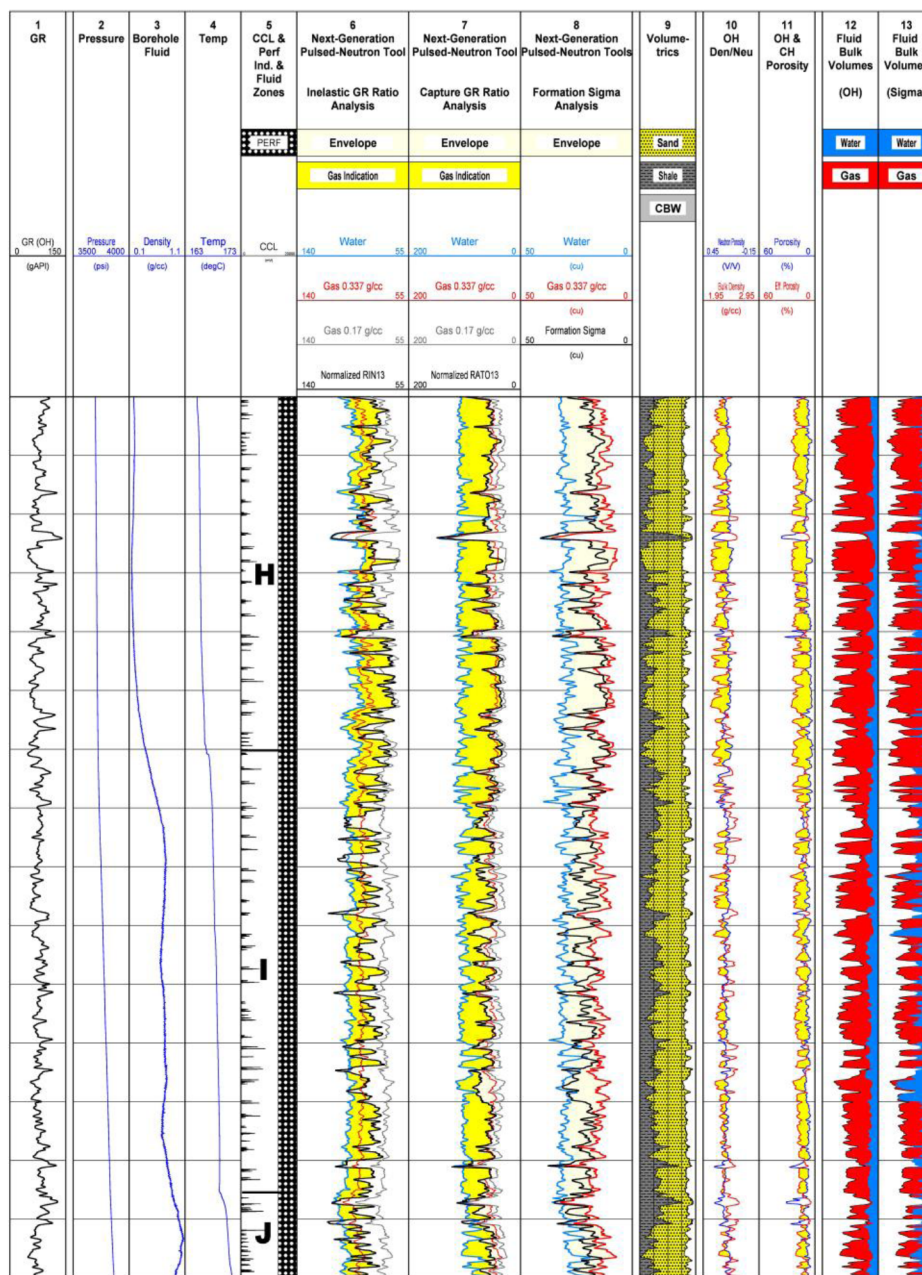


Figure 6—A composite plot presenting pulsed-neutron well logging data analysis for the lower sandstone formation. The presented interval was perforated.

The leftmost track presents an openhole gamma-ray log. The borehole pressure measurement is shown in Track 2. The borehole fluid density, calculated using measured pressure data, is plotted in Track 3. The pressure data analysis identified gas, gas-condensate-water mixture, and water columns with transition zones demarcating each column. Three distinct zonations were established to capture the variations in borehole fluid for RIN13 and RATO13 envelopes shown in Tracks 6 and 7, with zones marked on Track 5 as ‘H’, ‘I’, and ‘J’. Track 4 displays the borehole temperature profile, which closely mirrors the borehole pressure log profile as the borehole fluid type changes.

Tracks 6 and 7 present the RIN13 and RATO13 envelopes, with responses of 1.1 g/cc water and 0.337 g/cc gas (corresponding to the initial reservoir pressure), respectively. The analysis of reservoir depletion for the deeper sandstone formation was excluded because changes in water saturation, which serve as a reference in the analysis algorithm, occur as a result of production. Nevertheless, a preliminary qualitative

evaluation of reservoir pressure depletion was undertaken, based on the assumption that the water saturation obtained from the sigma analysis accurately reflects the in situ hydrocarbon saturation, thereby serving as a reference for the analysis of reservoir depletion. A liquid reinvasion effect on RIN13 could indicate less depletion than the actual reservoir pressure decrease, and the borehole fluid effect near the fluid transition zones impacts the RIN13-based analysis more significantly; therefore, RIN13 was not the primary choice for this zonal data analysis. The RATO13 analysis demonstrated that a gas density of 0.17 g/cc (shown as a gray line on Track 7) produced a gas saturation profile comparable to that obtained from the formation sigma analysis. Consequently, the estimated current reservoir pressure indicated a reduction of approximately 65% from the initial reservoir pressure, decreasing from approximately 15,000 psi to 5,000 psi at a formation temperature of 178 degrees Fahrenheit.

The formation sigma envelope and the measured data are shown in Track 8. The water saturation analysis, using formation sigma, revealed that the deeper formation contained less formation gas than original gas in place, except in specific zones where the lower portion of zone 'H' exhibited a similar level of gas to the initial gas volume. The original and current overlays of gas indication are presented in Tracks 10 and 11, respectively. The actual formation fluid volumes in the original and current (2024) are shown in Tracks 12 and 13, revealing the increase of water over time.

Summary and Conclusions

We performed time-lapse reservoir depletion monitoring for the North Sea offshore gas-condensate-bearing reservoirs. We presented the specifications of sleeved-pulsed-neutron well logging and data interpretation workflows in this paper. Furthermore, we emphasized the significance of monitoring gas reservoirs prior to perforation and presented the production outcomes attained through optimized perforation guided by this monitoring analysis.

The essential points are outlined as follows:

- The RIN13 and RATO13 envelopes were developed using MCNP models in conjunction with volumes of sand and shale, as well as the densities of water and various gases. The measured RIN13, RATO13, and formation sigma logs were then compared in each respective envelope to assess reservoir depletion for RIN13 and RATO13 or to calculate the current water saturation for formation sigma.
- The data obtained in 2024 using the next-generation pulsed-neutron tool showed a high degree of coherence with the previously acquired data recorded with the previous-generation pulsed-neutron tool. This finding confirmed a seamless transition from the legacy tool to the next-generation tool, thereby facilitating uninterrupted analysis of time-lapse data.
- The integration of pulsed-neutron well logging with an advanced reservoir pressure monitoring technique provided conclusive evidence of reservoir depletion. The upper sand bodies in the upper formation remained in a virgin reservoir state. In contrast, the lower sand bodies of the same formation, which were disconnected from the upper sand bodies, experienced reservoir depletion. The estimated reduction in reservoir pressure was approximately 45% of the initial reservoir pressure.
- The perforation of the target reservoirs has substantiated that the virgin reservoir pressure condition corresponded with the expectations established from the pulsed-neutron logging assessment. This demonstrates that the monitoring analysis provided vital insights for perforation decisions, particularly in situations where specific sands within the same formation had experienced depletion. Furthermore, it elucidates that the adjacent sand bodies were not interconnected, indicating that the reservoir pressure depletion did not transpire for all sand bodies within a single sandstone formation.

This surveillance project demonstrates the advancements in pulsed-neutron well logging technology, combined with an advanced data interpretation workflow, which provided the operator with an effective method to evaluate the reservoir behind casing in a non-destructive manner to the existing completion (including casing and cement sheath) and formation fluid dynamics (i.e., such as no disturbance of multiphase flow paths or the causation of crossflows between sand bodies). Ultimately, this technology enables strategic planning for enhancing well productivity.

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